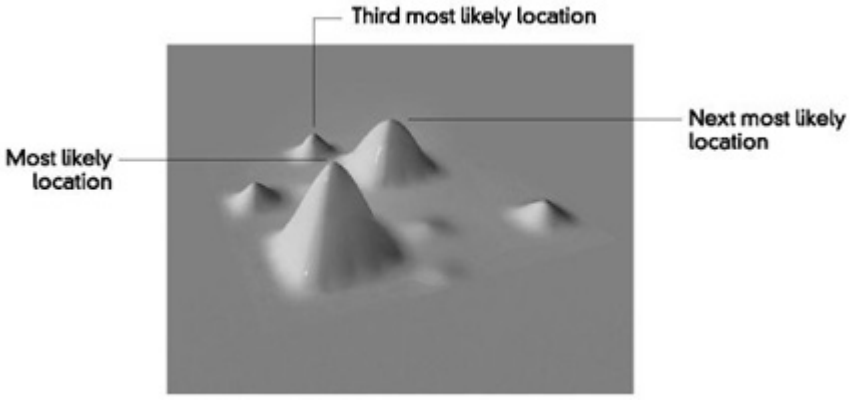


Figure 4.5 The probability wave of a particle, such as an electron, tells us the likelihood of finding the particle at one location or another.

Only a portion of an electron's probability wave is shown in Figure 4.5: according to quantum mechanics, every probability wave extends throughout all of space, throughout the entire universe.[6] In many circumstances, though, a particle's probability wave quickly drops very close to zero outside some small region, indicating the overwhelming likelihood that the particle is in that region. In such cases, the part of the probability wave left out of Figure 4.5 (the part extending throughout the rest of the universe) looks very much like the part near the edges of the figure: quite flat and near the value zero. Nevertheless, so long as the probability wave somewhere in the Andromeda galaxy has a nonzero value, no matter how small, there is a tiny but genuine—nonzero—chance that the electron could be found there.

Thus, the success of quantum mechanics forces us to accept that the electron, a constituent of matter that we normally envision as occupying a tiny, pointlike region of space, also has a description involving a wave that, to the contrary, is spread through the entire universe. Moreover, according to quantum mechanics this particle-wave fusion holds for all of nature's constituents, not just electrons: protons are both particlelike and wavelike; neutrons are both particlelike and wavelike, and experiments in the early 1900s even established that light—which demonstrably behaves like a wave, as in Figure 4.1—can also be described in terms of particulate ingredients, the little "bundles of light" called photons mentioned earlier.[7] The familiar electromagnetic waves emitted by a hundred-watt bulb, for example, can equally well be described in terms of the bulb's emitting about a hundred billion billion